

Akshay Dahiya

✉ xadahiya@gmail.com | in xadahiya | 🐦 @xadahiya

Experience

Blockchain Developer

Dec 2020 - Present

FREELANCER

- **Deployed 50+ Smart contracts** in production over multiple EVM compatible chains.
- Designed numerous **ERC20, ERC720, ERC720A, and ERC1155** contracts including hyper-deflationary and reward tokens for multiple projects on ETH, AVAX, BSC and Polygon.
- Developed multiple **Node projects** with features like dynamic reward rates, evolving NFTs, a referral system, and NFTs tradable on OpenSea.
- Designed multiple Trading and Sniper bots using Web3py and Solidity for NFT minting and Token trading.
- Developed a **Play2Earn** game using **custom Chainlink Oracle**, deployed on Binance Smart Chain and Polygon with a functional bridge.

Senior Software Engineer

April 2020 - Present

HYPERSONIX INC.

- Designed two critical services API Gateway and Management Platform from scratch for better performance and speedy development.
- **Improved API performance by about 80%** by code optimization, re-architecture of flow, and incorporating compression algorithm (Brotli).
- Designed a robust and secure authentication mechanism by combining Django Auth, API keys, OAuth, and SSO.
- Designed multiple components like logging and performance monitoring systems, a coding quality assurance framework, etc.

Master's Project & Special Problems

Jan 2020 - May 2021

GEORGIA INSTITUTE OF TECHNOLOGY

- Worked with **Dr. Thad Starner** to develop tools to Counter Plagiarism and assist online Education at Scale.
- Our research project Jack Watson got **featured in Forbes, TimesHigherEducation**, and multiple other places.
- Worked on 4 research projects on Counter Plagiarism and **published 2 papers at ACM Learning@Scale**.

Graduate Programmer

June 2018 - Dec 2018

FIDELITY INTERNATIONAL

- Developed a real-time survey and feedback WebApp using WebSockets, Django, Redis, and D3.js.
- Worked in the behavioral finance team and helped design tools for portfolio managers using React and Flask.

Founder

Aug 2015 - Dec 2017

TYPINGGEEK

- **Founded** Typinggeek, a platform to learn to type without wasting time on meaningless phrases using Django, Javascript, and Web scraping.
- Handled everything from User research to product development to final deployment.
- Built a user base of 5000+ Global Users.

Google Summer of Code Participant

May 2017 - Aug 2017

PYTHON SOFTWARE FOUNDATION

- **Designed Hydrus, a web framework built in Flask** that can create semantic web-based Restful APIs using a single specs file (Hydra Vocabulary)
- **Built a simulation (Flock demo)** where multiple servers and clients interact autonomously to demonstrate Semantic Web capabilities.
- The project has **188 Stars** and **135 Forks** (to date) on Github.

Volunteer Experience

Administrator, Google Summer of Code

Feb 2018 - Aug 2021

HYDRAECOSYSTEM.ORG

- **Mentored 6 students** for **Google summer of code** over a period of **3 years**.
- We developed tools to make the Hydra framework operational using Python and lower the Semantic Web entry barrier.

Mentor & Project Reviewer

Feb 2017 - Mar 2020

UDACITY

- **Assisted 1000+ students** with questions related to projects, content and Code for Full Stack and Deep Learning Nanodegrees.
- **Reviewed 5000+ projects** providing valuable insights to students related to better coding and design practices.

Skills

Languages: Python, Solidity, Javascript, C++(basic), Rust(basic)

Web: Django, Flask, FastAPI, PostgreSQL, Redis, ELK, Auth0, Celery, Websockets, Semantic Web

AI: Numpy, Pandas, Sklearn, OpenCV, Tensorflow, Pytorch, Fast.ai

Blockchain: Hardhat, Ganache, Truffle, Web3js, Web3py, Ethers, Chainlink Oracles, NFT, DAO, React

Other: Algorithms, Data Structures, Docker, AWS, GCP, Git, Linux, Inkscape, UX Research, UI Design, Unit testing, Web scraping

Education

MS in Computer Science

GEORGIA INSTITUTE OF TECHNOLOGY

- GPA 4.0/4.0 (12 CS research credits)
- Machine Learning Specialization

Jan 2019 - May 2021

Atlanta, USA

B.Tech in Electronics and Communication

DCRUST

- CGPA 7.05/10

July 2014 - Apr 2018

Murthal, India

Publications

1. Akshay Dahiya, Rocko Graziano, India Irish, and Thad Starner. 2021. **JackMarker with GitDown: A Framework to Counter Plagiarism at Scale**. In Proceedings of the Eighth ACM Conference on Learning @ Scale (L@S'21). Association for Computing Machinery, New York, NY, USA, 335–338. DOI: <https://doi.org/10.1145/3430895.3460982>
2. Raghav Apoorv, Akshay Dahiya, Uma Sreeram, Bharat Rahuldhev Patil, India Irish, Rocko Graziano, and Thad Starner. 2020. **Examinator: A Plagiarism Detection Tool for Take-Home Exams**. In Proceedings of the Seventh ACM Conference on Learning @ Scale (L@S '20). Association for Computing Machinery, New York, NY, USA, 261–264. DOI: <http://dx.doi.org/10.1145/3386527.3406723>